

Final Project: Initial Analysis Report

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Introduction

In the intention of understanding the effect of Pitcher's stuff and regular performance metrics on the IL placement, we set the groundwork for the dataset being analyzed by effectively merging the required metrics and pitching stuff characteristics into one primary dataset that encompasses all pitchers being accounted for with a Statcast threshold, limiting the data to 2015-2023 statistics and placing a minimum data threshold of at least 2,000 pitches thrown in the course of these nine seasons. This is an optimum sample size, which, when combined, yielded a robust dataset of 826 observations with 43 variables, consisting of each qualified pitcher's statistical compilation across these nine seasons.

The combination of this dataset was performed using the Savant API system, which delivered a CSV, created with specific characteristics that included all relevant Statcast pitching metrics. Then, using R Studio, this CSV was effectively combined with the baseball R built-in library (Lahman), which provides a Pitching dataset table with statistics from 1871 to date, as indicated in the proposal/dataset selection deliverable. Using Normalizer helpers to standardize the names, both datasets were combined by matching names in common and removing all names not included in the Savant dataset. We also filtered the Lahman Pitching dataset to the intended range starting in 2015. Since this version of the library is limited to 2023, an end date was not required. We also began by creating interesting per 9 innings metrics, which were further explored through the EDA analysis, as fully explained in the following Analysis section. However, one of the most critical aspects of efficiently laying the groundwork for this data was

to check for data duplicates when creating the `lahman_pitcher_totals` data frame, which already included a cumulative statistical compilation of qualified pitchers.

At this moment, it was key to open the CSV separately and manually search for all duplicates to determine if they were present in the Savant dataset. To optimize the identification process of duplicate pitchers with the same names, we accessed TruMedia, which provides baseball teams with robust tools for data analytics, visualizations, and video scouting. Here, we could easily identify the players with their MLB IDs and remove those present in the Lahman Pitching library but not in Savant. Upon manually proceeding with this much-needed adjustment, we then loaded the Savant dataset, only including relevant and interesting variables and by immediately standardizing the names to make them match those in Lahman Pitching dataset, further checking for dupes which undelved a completely different process due to the duplicate of a pitcher named Luis Garcia which was the only one that actually qualified as a dupe due to his inclusion and qualification based on the Savant dataset. In this scenario, separate data frames were produced by merging the different Luis Garcia (one with an accent and the other without) across the various datasets. This was performed by mapping these names to distinct player IDs and then combining them with the rest of the already merged set.

All this, then permitted us to delve deeper into the data and understand regular data behavior through the use of practical EDA analysis which incorporated correlation matrix, correlation heatmaps, histograms, scatterplot, density plots and linear regression models that will enhance the interpretability of this initial analysis in means of preparing it for further implementation of IL statistics currently ranging from 2015-2017, per publicly available data, which will permit us to discover hidden trends and project the statistical performance of this IL factored binary outcome for subsequent years!

Methods

The two datasets (Lahman and Savant), as mentioned previously, were combined in R into one clean dataset called `pitcher_df` to ensure that player records were normalized and combined by name from both datasets. As previously mentioned, the data was restricted to the Statcast era (2015-2023) to guarantee the availability of modern pitching metrics. Player performance metrics were also aggregated at the career level to enable analysis of long-term pitching outcomes.

For every pitcher who appeared from 2015 to 2023, individual season statistics from the Lahman Pitching dataset were grouped and summed across seasons. This included career innings pitched, career earned run average, career strikeouts per nine innings, career walks per nine innings, career average homeruns allowed per nine innings, and career walks plus hits per nine innings. This career-level aggregation enabled merging with the Statcast data (velocity, spin rate, etc.), providing a complete dataset that merges every metric for every pitcher. This dataset included one record for each pitcher, enabling the analysis of relationships between performance metrics and overall results, such as career ERA and total innings pitched, which will be further explored later in the Analysis section.

This analysis employs two methods for now but will use a third method (logistic regression) for further study. An exploratory data analysis was performed to understand better the dataset, including the distribution and relationships among the key variables. This included summary statistics, a correlation matrix, and different visualizations. Two multivariable linear regression models were also developed; one to model career ERA (continuous variable) and the other to model innings pitched as the continuous variable. These models allowed the evaluation

of both performance and longevity outcomes using the same set of predictors. Let us gain a better understanding of pitching effectiveness and durability.

Analysis

Exploratory Data Analysis (EDA)

After combining and cleaning the datasets, the data was prepared for exploration. An exploratory data analysis was conducted to get more insight into the data. As mentioned previously, the `pitching_df` dataset (combined dataset) contained over 826 observations with 43 variables. A summary statistic of key variables such as ERA career, SO9 career (strikeouts per nine innings), BB9 career (walks per nine innings), WHIP career (walks plus hits per nine innings), velocity, spin rate, and total pitches were analyzed (see Figure 1).

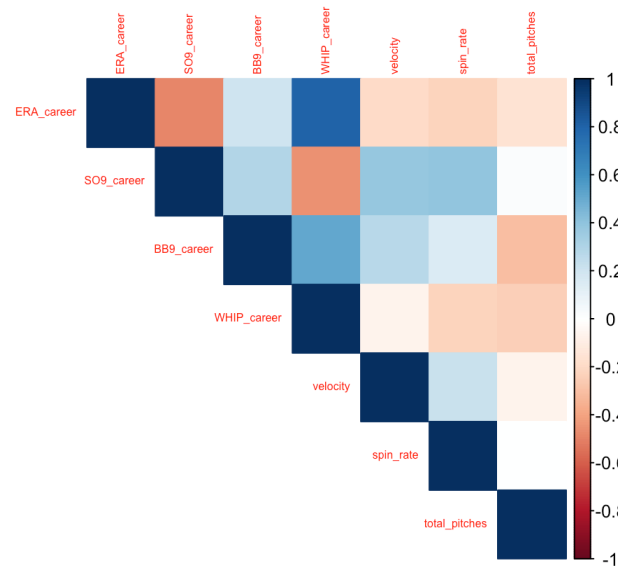
Figure 1

ERA_career	SO9_career	BB9_career	WHIP_career	velocity	spin_rate	total_pitches
Min. :1.670	Min. : 3.844	Min. :1.075	Min. :0.8923	Min. :75.00	Min. :1649	Min. : 2001
1st Qu.:3.638	1st Qu.: 7.366	1st Qu.:2.714	1st Qu.:1.2229	1st Qu.:87.22	1st Qu.:2097	1st Qu.: 3006
Median :4.185	Median : 8.479	Median :3.258	Median :1.3165	Median :89.00	Median :2219	Median : 4604
Mean :4.194	Mean : 8.549	Mean :3.316	Mean :1.3140	Mean :88.90	Mean :2223	Mean : 6266
3rd Qu.:4.656	3rd Qu.: 9.630	3rd Qu.:3.819	3rd Qu.:1.4026	3rd Qu.:90.78	3rd Qu.:2337	3rd Qu.: 7798
Max. :7.000	Max. :14.744	Max. :6.021	Max. :1.7791	Max. :97.00	Max. :2778	Max. :25820

This table revealed interesting information about key pitching metrics in this dataset, such as the average career ERA being around 4.19, with the lowest average being 1.67. The average velocity seemed to be roughly 88.90 mph, and the total pitches averaged by pitchers seemed to be 6,266 pitches, which doesn't seem like a whole lot. Still, we must consider that these are career totals, and some pitchers have probably been in the league longer than others since the data is from the seasons 2015 to 2023.

A correlation matrix and correlation plot were produced using the same key variables as in the descriptive summary statistics table, as these variables are relevant to performance and pitching metrics. The correlation matrix revealed some meaningful relationships (see Figure 2).

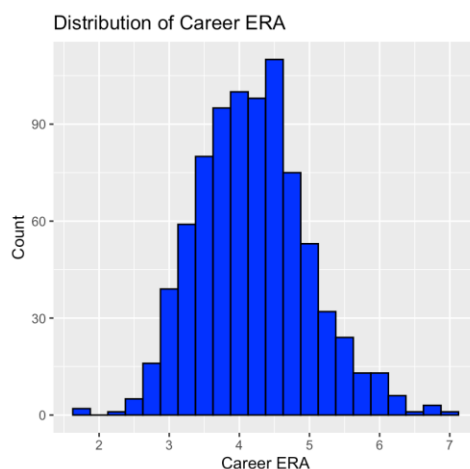
Figure 2



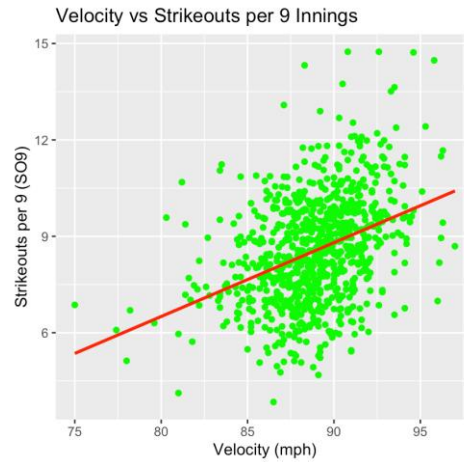
It seems that ERA had a strong positive correlation with WHIP ($r \approx 0.81$), which is expected, suggesting that pitchers who allow more baserunners also allow more runs. Strikeout rate (SO9) had a moderate negative correlation with ERA career ($r \approx -0.49$), indicating that more dominating pitchers with more strikeouts generally have lower ERAs. SO9 also had a mild positive correlation with both velocity ($r \approx 0.38$) and spin rate ($r \approx 0.40$), which also suggests that pitchers with greater velocity and spin rates tend to produce more strikeouts. Walk rate (BB9) was moderately correlated with WHIP ($r \approx 0.52$) and had a weak positive correlation with ERA ($r \approx 0.21$), highlighting the influence of control on performance metrics. Overall, these correlations back the theoretical assumption that higher “stuff” metrics lead to better results. Visualizations also supported these correlations.

A histogram of the distribution of career ERA among pitchers was created (see Figure 3). The histogram supported the summary statistics table, showing a slightly right-skewed distribution, with many pitchers having an ERA between 3.0 and 5.0. Some outliers exceeded 5.0, indicating poorer performances for those pitchers. This distribution suggests that although most pitchers perform better or close to the league average, a few underperform, which increases the spread of the data.

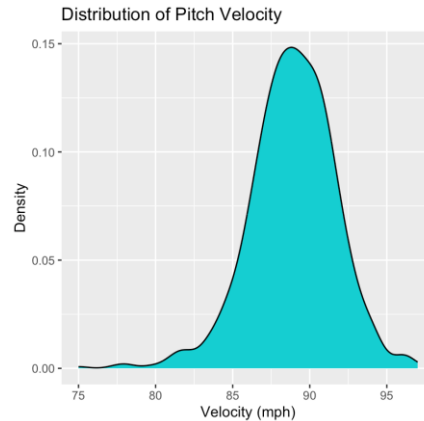
Figure 3



A scatterplot was also created to look at the relationship between velocity and SO9 (see Figure 4). The correlation plot showed a moderate positive correlation between the two, and the scatterplot supported this finding. The scatterplot showed a positive linear relationship between the two. As velocity increases, so does strikeout rate, indicating that pitchers who throw faster and harder tend to strike out more batters. The linear trend line confirms a moderate positive correlation.

Figure 4

Lastly, for the EDA, a density plot of velocity was also created (see Figure 5). The density plot shows the distribution of average pitch velocity among the pitchers in the dataset. The plot showed a left-skewed distribution, suggesting that most pitchers in the dataset throw at relatively high pitch velocities. In contrast, a smaller group of pitchers throw at significantly lower speeds. Most velocities were focused around 90 mph, as shown in the descriptive statistics table, which indicated that the average velocity was roughly 88.90 mph. This skew also suggests that while high-velocity pitching has become more common in the MLB, there exists a group of pitchers whose success could rely on factors other than just velocity. This also provides potential information on how this could influence injury risks since higher velocities could put more stress on the pitcher's body. However, even though, this could be used for further analysis when applying a logistic regression model on injury outcomes it is also worth mentioning that the observed average speed and other pitch type related metrics might be slightly altered as they include a combination of all pitch types which consists of varying pitch designs, spin rates and velocity (FB, CH, SL, CB, etc...).

Figure 5

Linear Regression Models

To quantitatively assess the relationships identified in the EDA, two multivariable linear regression models were developed. These models aim to predict key outcomes, pitcher effectiveness (Career ERA), and durability (Total Innings Pitched), using a set of core performance and "stuff" metrics.

Model 1: Predicting Career ERA

The first model (`lm_era`) was built to predict a pitcher's career Earned Run Average (`ERA_career`) based on their career strikeouts per 9 innings (`SO9_career`), walks per 9 innings (`BB9_career`), average pitch velocity, and `spin_rate`.

Figure 6

```
Call:
lm(formula = ERA_career ~ SO9_career + BB9_career + velocity +
    spin_rate, data = pitcher_df)

Residuals:
    Min       1Q   Median       3Q      Max
-2.06657 -0.39430 -0.02311  0.38127  2.40404

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.6022032   0.7402143   10.270  <2e-16 ***
SO9_career   -0.2603345   0.0151033  -17.237  <2e-16 ***
BB9_career    0.3666996   0.0268436   13.661  <2e-16 ***
velocity     -0.0219823   0.0085744   -2.564   0.0105 *
spin_rate     -0.0001996   0.0001322   -1.510   0.1315
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6185 on 821 degrees of freedom
Multiple R-squared:  0.3797,    Adjusted R-squared:  0.3767
F-statistic: 125.6 on 4 and 821 DF,  p-value: < 2.2e-16
```

The model is statistically significant (F-statistic - p-value < 2.2e-16) and explains approximately 37.7% of the variance in `ERA_career` (Adjusted R-squared = 0.3767). For each

additional strikeout per 9 innings, a pitcher's career ERA is predicted to decrease by 0.26, holding other factors constant. This is a highly significant predictor, confirming that dominant, high-strikeout pitchers are more effective at preventing runs. For each additional walk per 9 innings, a pitcher's career ERA is predicted to increase by 0.37. This highly significant result underscores that poor control is hugely detrimental to a pitcher's success. It also permitted us to determine that for each 1 mph increase in average velocity, ERA is predicted to decrease by 0.022. This finding is statistically significant ($p = 0.0105$) and supports the idea that higher velocity contributes to better run prevention. Ultimately, the spin rate was not deemed a statistically significant predictor of ERA ($p = 0.1315$) when controlling the other variables.

Model 2: Predicting Total Innings Pitched (Durability)

The second model (`lm_ip`) was designed to predict a pitcher's total career innings pitched (`IP_total`), serving as a proxy for durability and long-term workload capacity.

Figure 7

```
call:
lm(formula = IP_total ~ SO9_career + BB9_career + velocity +
    spin_rate, data = pitcher_df)

Residuals:
    Min       1Q   Median       3Q      Max
-490.44 -184.76  -77.19  113.26 1230.97

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  530.85068    339.52150     1.564 0.118314
SO9_career    24.02756     6.92759     3.468 0.000551 ***
BB9_career  -141.73000    12.31261   -11.511 < 2e-16 ***
velocity      0.75902     3.93291     0.193 0.847013
spin_rate     0.03586     0.06065     0.591 0.554522
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 283.7 on 821 degrees of freedom
Multiple R-squared:  0.1428,    Adjusted R-squared:  0.1386
F-statistic: 34.19 on 4 and 821 DF,  p-value: < 2.2e-16
```

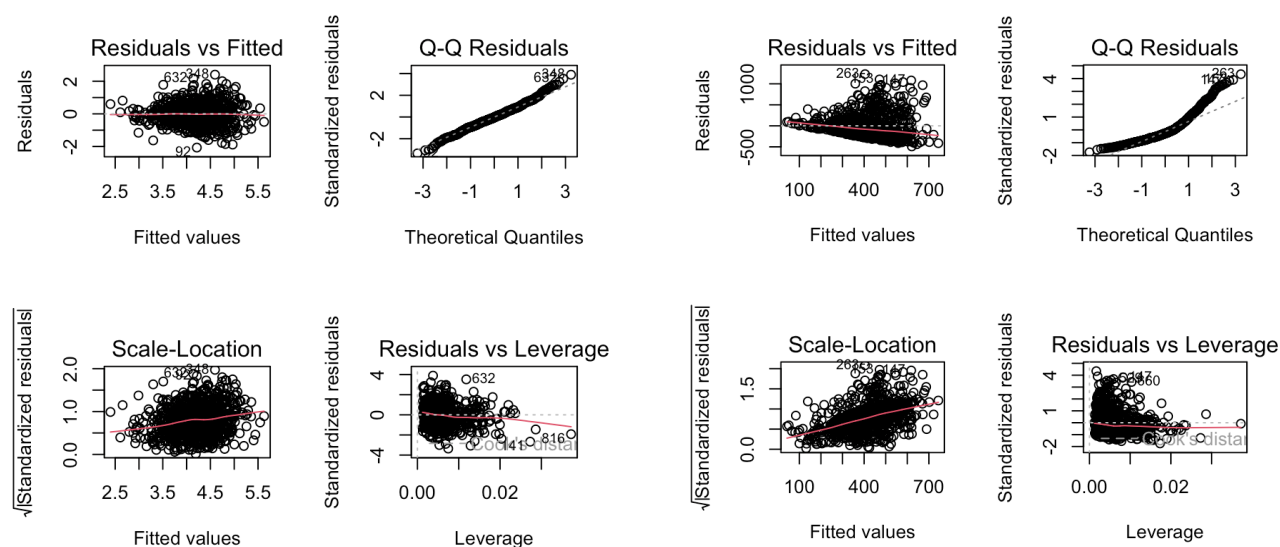
While the overall model is statistically significant ($p\text{-value} < 2.2e-16$), it explains only 13.9% of the variance in `IP_total` (Adjusted R-squared = 0.1386). This suggests that these specific metrics are not strong predictors of a pitcher's total career workload on their own. Pitchers with higher strikeout rates tend to accumulate more innings over their careers, likely because their effectiveness keeps them in pitching rotations, permitting us to identify it as a

significant positive predictor. During our model evaluation, we observed that BB9_career was the strongest predictor, exhibiting a significant negative relationship. This indicates that for every additional walk per 9 innings, a pitcher is expected to throw approximately 142 fewer career innings. This suggests that pitchers with poor control receive far fewer opportunities over the long term. Lastly, regarding velocity and spin rate, neither of these "stuff" metrics was found to be a statistically significant predictor of the total innings a pitcher throws.

Regression Diagnostics

To ensure the validity of the linear regression models, diagnostic plots were generated to check for violations of key statistical assumptions: linearity, normality of residuals, homoscedasticity (constant variance), and the presence of influential outliers.

Figure 8 and Figure 9



The Residuals vs. Fitted plot showed a reasonably random scatter of points around the horizontal line at zero for the ERA model (Figure 8), indicating that the linearity assumption is reasonably satisfied. The plot for the IP model (Figure 9) reveals a slightly downward trend, suggesting that as fitted values increase, residuals decrease. This indicates non-linearity, and that

model does not fully capture the relationship between the predictors and innings pitched. The spread of residuals also seems to increase when the fitted values change, indicating heteroscedasticity. On the other hand, the Normal Q-Q Plot points fell closely along the diagonal dashed line, suggesting that the residuals are approximately normally distributed, satisfying a key assumption for inference for the ERA model. There are a few outliers on the upper tail and lower tail, but they are not severe enough to significantly affect model validity. The Q-Q Plot for the IP model, however, indicated that residuals diverge considerably from the diagonal line, violating the assumption of normality and suggesting that the residuals are not normally distributed. Extreme outliers can also be seen in the plot.

The Scale-Location Plot for the ERA model displayed a relatively consistent spread of standardized residuals across fitted values, with a slight upward trend visible in the red line. This suggests some increase in residual variance for higher fitted ERA values, but the impact seems limited. The model typically meets the assumption of constant variance. The plot for the IP model displays a clearer upward slope in the red line trend, which shows an increase in residual variance as fitted values increase. This confirms heteroscedasticity as the model predicts the number of innings pitched with less accuracy for pitchers who log more innings. The final plot, the Residuals vs. Leverage Plot, helps identify influential outliers. The plot for the ERA model reveals that most points are clustered within acceptable leverage limits, suggesting that no extreme outliers have exerted disproportionate influence. While a few observations had higher leverage (632 and 816), none fell outside of Cook's distance, indicating no single data point was unduly driving the regression results. The plot for the IP model displays several points with moderately high leverage that might have a disproportionate influence on the regression results.

However, most points lie within the acceptable ranges of Cook's distance, suggesting no severe outliers.

Overall, the diagnostic plots of the linear regression model predicting ERA based on SO9, BB9, velocity, and spin rate perform reasonably well. Residuals seem predominantly random; no extreme influential points were detected. Therefore, the model's assumptions are largely satisfied, and the fitted model can be considered reliable for inference and prediction. The diagnostic plots for the linear regression model predicting total innings pitched based on SO9, BB9, velocity, and spin rate display certain violations of linear regression assumptions. The Residuals vs. Fitted Plot and Scale-Location Plot especially reveal non-linearity and heteroscedasticity. At the same time, the Q-Q Plot also shows slight departures from normality in both the lower and higher tails. These trends suggest that pitchers experiencing excessively high workloads are not well represented by this linear model. Transforming the dependent variable into $\log(\text{IP_total})$ might improve model fit.

Logistic Regression for Injury Risk

The linear regression models provided valuable insights into performance and durability, but the project's central aim is to understand injury risk. Since placement on the Injured List (IL) is a binary outcome (a pitcher either is or is not placed on the IL), a different modeling approach is required. For this reason, the next phase of the analysis will employ logistic regression.

This model will be structured to predict the probability of an adverse health event, with a binary dependent variable (IL_Placement), obtained from future data to be incorporated. This model will also count on several predictors (independent variables), which might include "stuff" metrics (velocity, spin_rate, etc.), workload metrics (total_pitches, IP), and performance metrics (SO9_career, BB9_career, etc.). Unlike linear regression, which predicts a continuous value,

logistic regression models the probability of a specific outcome occurring. The results will be interpreted using odds ratios, which will quantify how a one-unit change in a predictor (e.g., a 1 mph increase in velocity) impacts the odds of a pitcher getting injured. This advanced method will allow us to directly test the hypothesis that elite "stuff" and high workloads increase the likelihood of injury.

Conclusion

This initial analysis report successfully uncovered hidden trends in pitching strategies and performance metrics, providing a strong foundation before incorporating the IL data into this analytical research study. Using successful merging techniques thoroughly described in the introduction of this paper, exploratory data analysis, and optimum analytical procedures, we were able to develop a robust structure that will serve as a driving force for analysts and teams trying to maximize the use of pitchers' arms and stuff composition during high leverage situations, while keeping them healthy.

Preliminary results suggest that detrimental factors affecting pitcher longevity in today's game may include elevated walk rates (BB9) and potentially excessive workloads. Conversely, higher strikeout ability and controlled velocity appear to be beneficial to effectiveness, though not consistently predictive of long-term durability. These findings indicate that while velocity remains a performance asset, balancing it with control and workload management could be critical for sustaining pitcher health and longevity.

This analysis serves as a crucial foundation for the project's upcoming phase, where we intend to implement a logistic regression model to predict the probability of IL placement. Our primary goal is to connect how pitching "stuff" predicts injury risk. We aim to achieve this by

integrating publicly available IL data and potentially enhancing it through MLB contacts, whom we are trying to reach, to aggregate missing data years for IL purposes (2018-2023) rather than just predicting these years' behavior. The evolution of this work frame will allow us to explore additional hidden trends connecting performance, workload, and durability. It offers a data-driven perspective on one of the most pressing challenges in baseball today, directly testing whether the modern emphasis on high velocity and spin rates comes at the cost of pitcher health.

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